

FOLLITROPIN ALFA -> 46,XX gonadal dysgenesis

For the attention of: A Dierich, Centre National de la Recherche Scientifique

Why this hypothesis is real

Drug-target convergence on the disease scores **0.86** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Therapeutic direction is **aligned** with OT genetic evidence (the drug pushes the target the disease-protective way). Lead target is well expressed in the disease-relevant tissue (ovary). Orthologous-gene model-organism evidence is strong (phylo_score 0.83, n=10).

Why now (IP runway + literature footprint)

The substance is a biologic or otherwise outside the FDA Orange Book; check the Purple Book and direct registry for IP runway.

Clinical opportunity

US prevalence figures are not in the curated map; for an orphan condition this is expected. The KOL can size the addressable population.

Why you

You were identified from a PubMed pass over recent papers co-mentioning the target and the indication (1 relevant PubMed publication).

What we propose

BioBix (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:

- Target validation across blood / immune / cancer multi-omics datasets
- Biomarker candidate identification from public + private cohorts
- Patient stratification analytics for trial-enrichment design
- Mechanism-of-action mining via systems-biology priors
- Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

- Investigator-initiated trial framework with the substance owner
- BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
- Joint publication strategy; data-sharing agreement gated on IP terms

Next step

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

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